pathologist Albert Frank in 1885. Studying what he first assumed was a pathological infection, Frank discovered that trees with the fungi attached to their roots were healthier than those without. The fine filaments, or hyphae, of the fungi make the roots more efficient at obtaining nitrate and phosphate nutrients from the soil. In return, the fungi get sugar and carbon from the tree.

Connected lives

No organism lives in isolation from the rest of its ecosystem. The behavioral interactions between them are complex and much is still being discovered about them. One of the greatest contributions in this field was made by British zoologist Charles Elton, whose 1927 classic *Animal Ecology* established many important principles of animal behavior, including food webs and food chains, prey size, and the concept of ecological niches.

Ethology, which looks at animal behavior and its evolutionary basis and development, is a major component in the modern study of organisms. Back in 1837, British entomologist George Newport discovered that moths and bees could raise the temperature of their thorax by quivering their muscles. From the 1970s onward, German-American entomologist Bernd Heinrich and others uncovered more thermoregulatory adaptations that have helped insects thrive. As heterotherms, they are able to maintain different temperatures in different parts of the body.

Modern research now combines laboratory experiments, field observation, and new technology such as infrared thermography to understand insect behavior in ever more detail.

Field observations are a key tool in ethological research. In the 1940s, British ornithologist David Lack investigated the factors controlling the number of eggs birds laid (clutch size). His food limitation hypothesis states